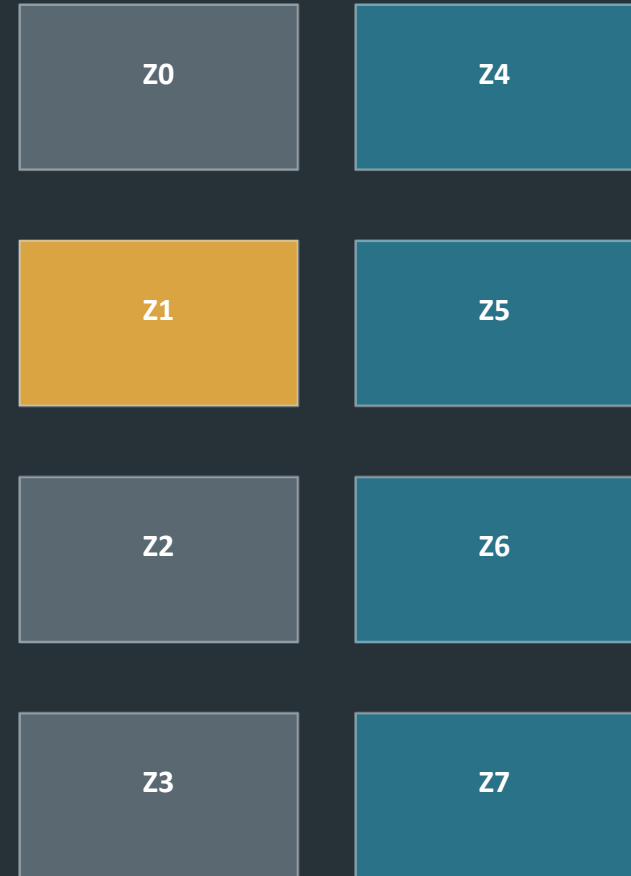


STATIC ZONES OR FREE SHUTTLE MOBILITY?

Characterizing isolation, locality, and coordination in scalable glass storage



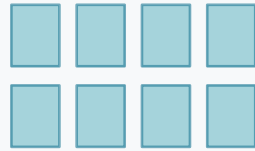
Capacity Scales, Active Resources Do Not

More passive media increases transport work without adding service bandwidth.

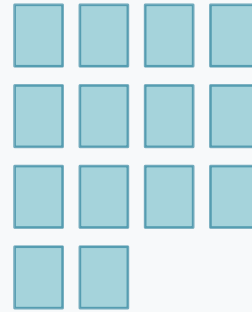
CAPACITY



Small



Medium



Large



FIXED ACTIVE RESOURCES

8

SHUTTLES

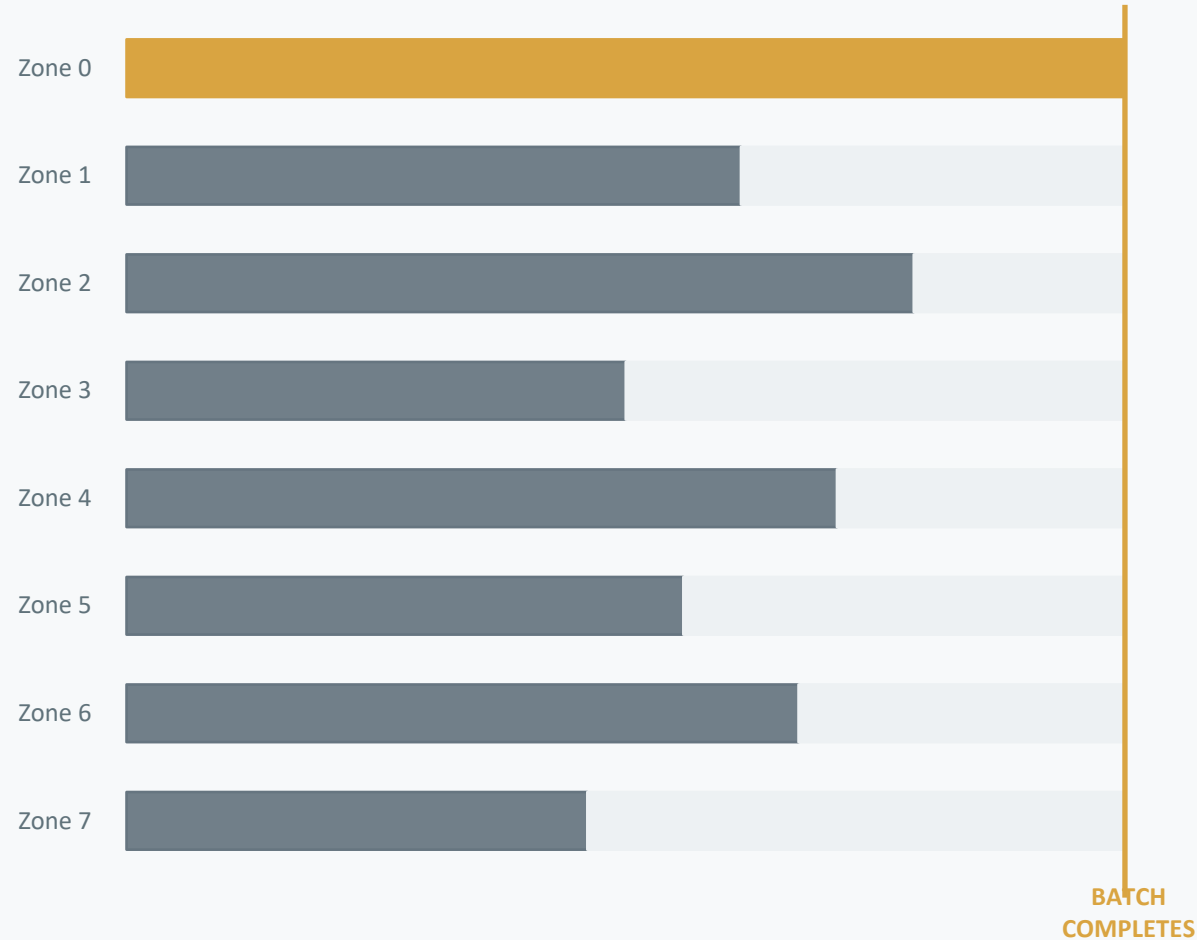
8

READERS

Question: can fixed transport and read resources preserve performance as the library grows?

Initial Motivation: The Slowest Zone Sets the Tail

Static ownership creates independent queues that cannot share shuttle capacity.



Potential cost

- Queued requests remain in one zone
- Other shuttles finish early
- Idle capacity cannot help the tail

Stranded service capacity

What Did We Want to Test?

Does shared mobility convert idle capacity into lower tail latency?

HYPOTHESIS 1

**No-Zone reduces
idle shuttle capacity.**

All shuttles draw work from a shared pool.



HYPOTHESIS 2

**Lower idle capacity
reduces tail latency.**

More active shuttles should drain work sooner.

Two Baselines

The comparison isolates fixed ownership versus shared movement.

STATIC ZONE

Zone 0 → Shuttle 0

Zone 1 → Shuttle 1

Zone 2 → Shuttle 2

Zone 3 → Shuttle 3

- Fixed owner
- Fixed reader
- Local movement
- Conflict-free abstraction

GREEDY NO-ZONE

Oldest task

Nearest idle shuttle

Nearest reader

Reserve route

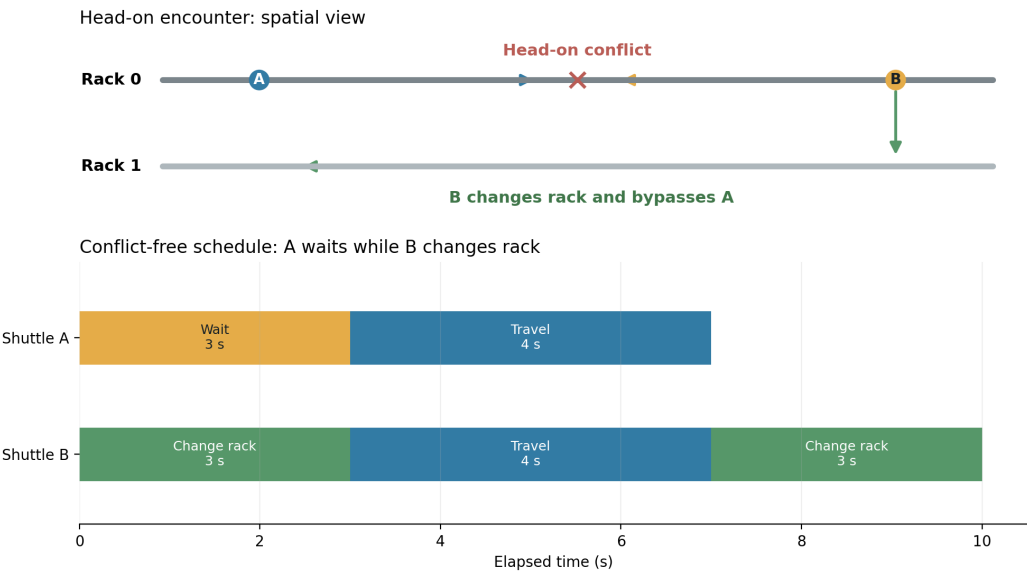
Shared work pool

Freedom is useful only when scheduling converts it into productive work.

How Greedy No-Zone Handles Conflicts

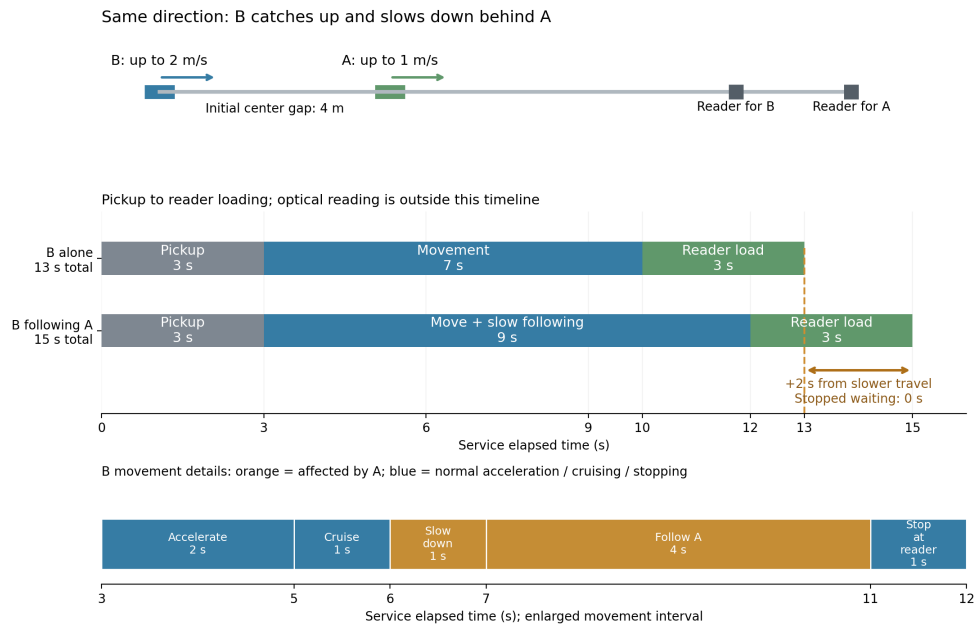
Direct routes are delayed or detoured when shuttle trajectories overlap.

HEAD-ON ENCOUNTER



B changes rack; A waits until the original rack is clear.

SAME-DIRECTION CATCH-UP



Cost accounting: baseline work 13/15 = 86.7%; slowdown penalty 2/15 = 13.3%; stopped waiting = 0%.
Illustrative motion model with separate available readers; slowdown penalty is a time difference, not an idle interval.

The following shuttle slows or waits to preserve clearance.

Trace-Driven Experiment

Both policies receive the same requests and physical placement.

100,000

Azure blob requests

640

Virtual platters

8 + 8

Shuttles and readers

10

Paired placement seeds

PANEL-SIDE LENGTH



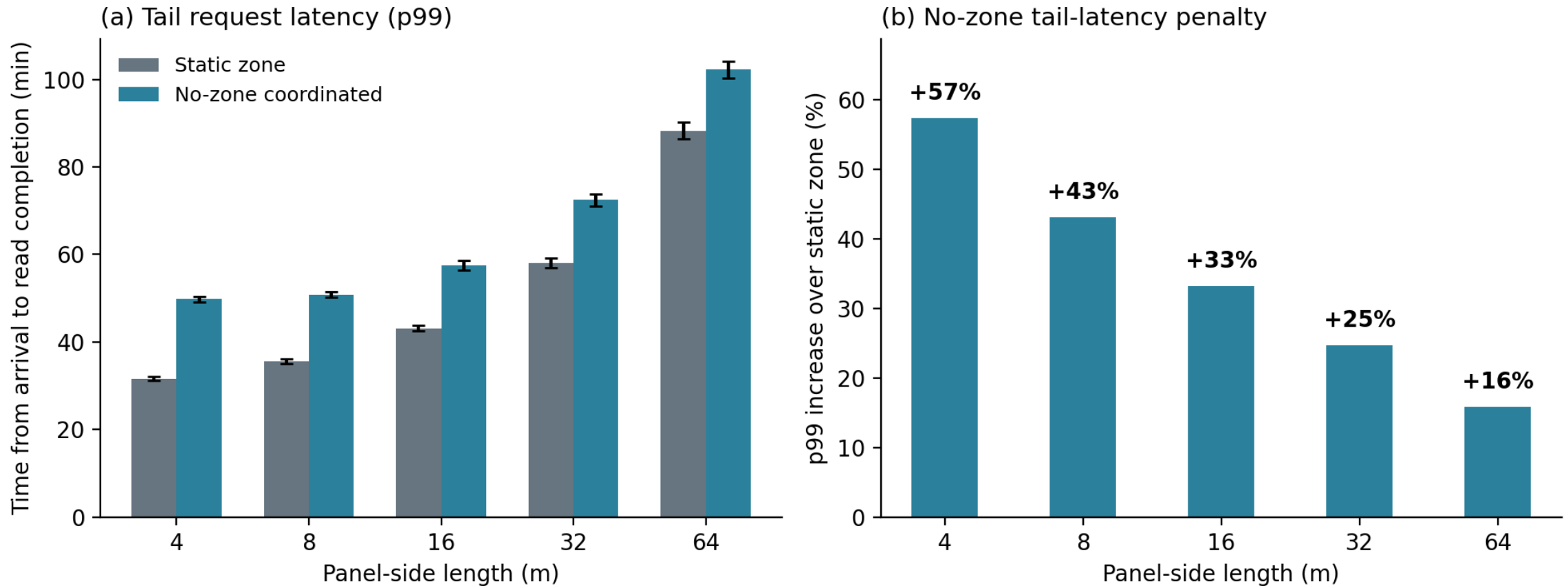
Preserved from the raw trace

Arrival time · request order · object identity · bytes

Physical glass locations are seeded synthetic mappings shared by both policies.

Unexpected Result: Greedy No-Zone Is Slower

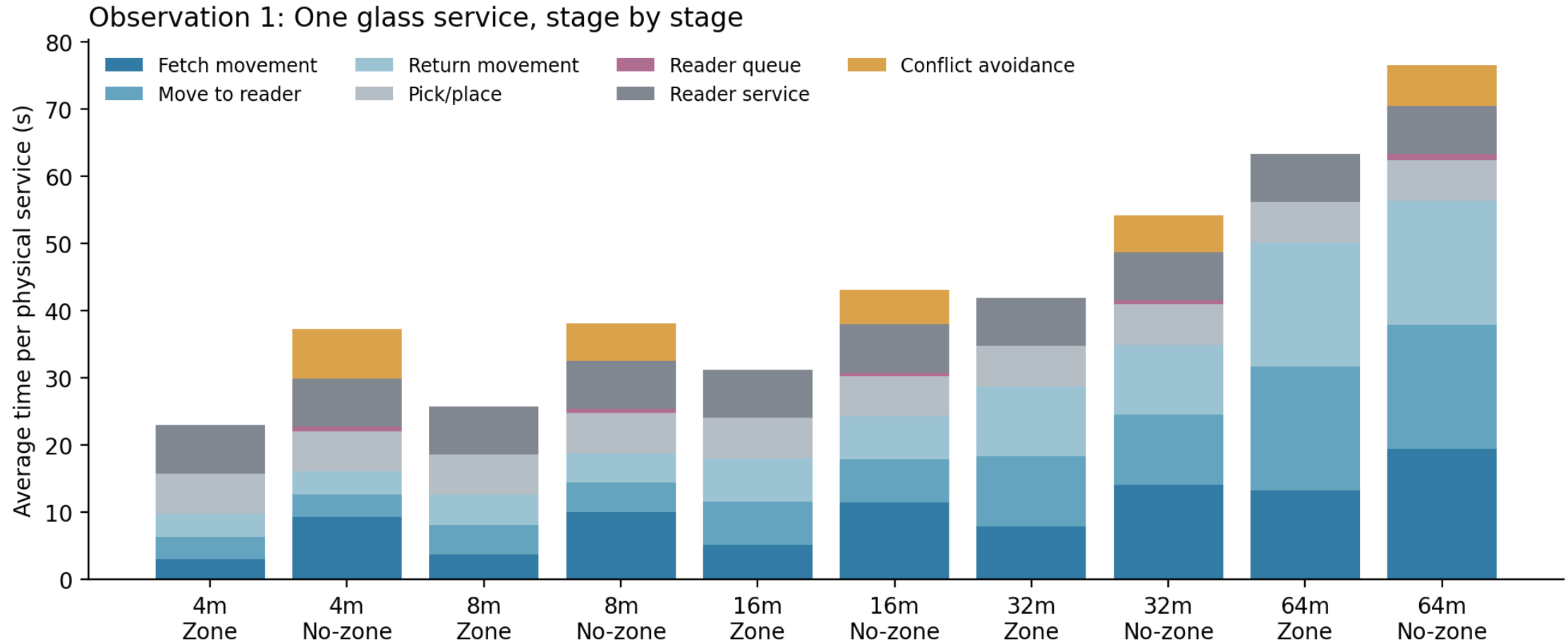
Shared mobility lowers idle capacity, but p99 request latency remains higher.



Same Azure arrivals and placement. Abstract holding-pocket model; error bars show placement-seed SD.

Why Is Greedy No-Zone Slower?

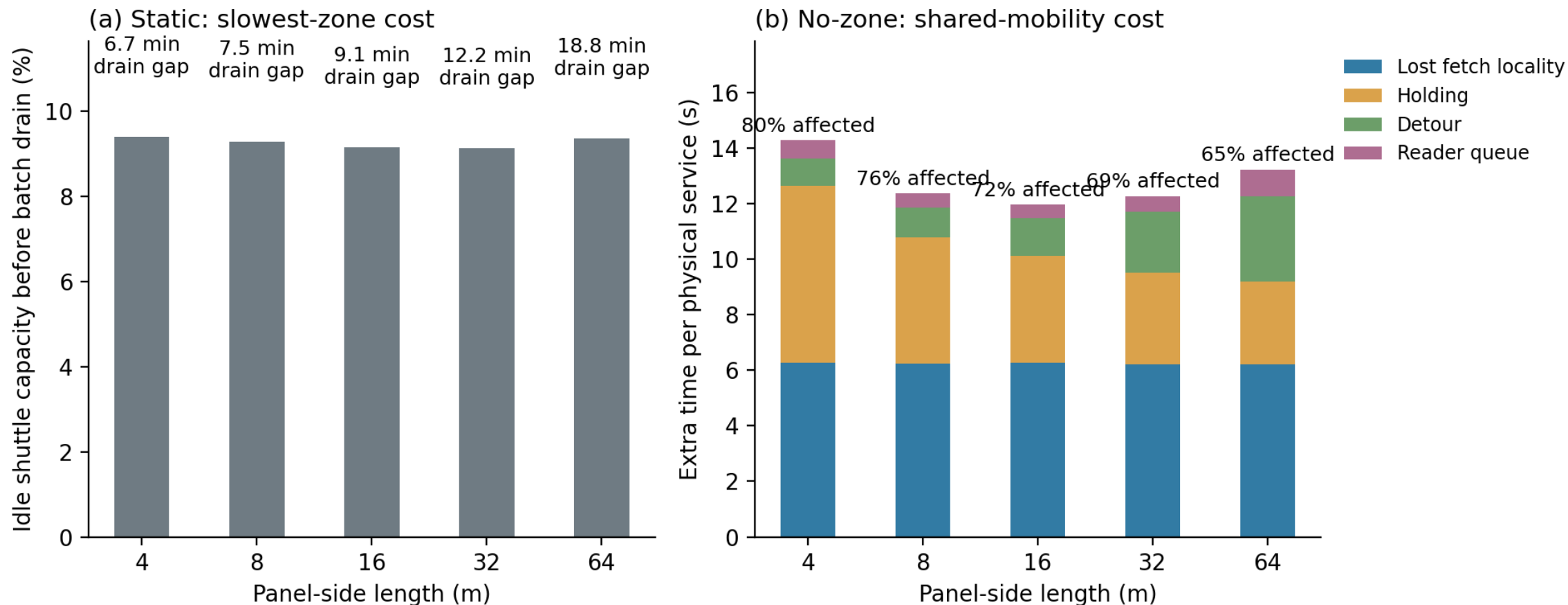
Each physical glass service becomes less efficient.



Same 640 batch-merged platter tasks under both policies; request arrivals are held behind a batch barrier.

The Real Trade-off

Static pays in stranded capacity; No-Zone pays per service.



Static pays in stranded capacity; no-zone pays in lost locality, conflict avoidance, and shared-reader waiting.

Higher shuttle utilization does not imply higher productive throughput.

Recover stranded capacity without sacrificing locality and traffic isolation.

1

Fair baseline

Allow the flexible scheduler to emulate Static Zone.

2

Locality-aware assignment

Choose task, shuttle, and reader by expected completion.

3

Bounded coordination

Plan only interacting shuttle paths in a short horizon.

4

Feeder buffering

Overlap platter transport with reader service when starvation dominates.

Zone provides more than collision avoidance: it also preserves movement locality.